



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

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Div: CSE A1

Batch:2025-29

ASSIGNMENT No : 11

TITLE :Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Outcome

The program successfully defined an interface with an abstract method and implemented it in two separate classes, each providing its own version of the method. Execution confirmed that both classes correctly followed the interface contract while behaving differently.

Conclusion

Interfaces enforce a common structure across unrelated classes without dictating implementation, enabling multiple inheritance of type and improving flexibility and modularity in Java applications.

Exercise

1. Create an interface **Printable** and implement it in Student and Employee classes.

```
1  interface Printable {
2      void print();
3  }
4  class Student implements Printable {
5      String name = "Amit";
6      public void print() {
7          System.out.println("Student Name: " + name);
8      }
9  }
10 class Employee implements Printable {
11     String name = "Sneha";
12     public void print() {
13         System.out.println("Employee Name: " + name);
14     }
15 }
16 public class PrintableDemo {
17     Run | Debug
18     public static void main(String[] args) {
19         Printable p1 = new Student();
20         Printable p2 = new Employee();
21         p1.print();
22         p2.print();
23     }
24 }
```

```
Student Name: Amit
Employee Name: Sneha
```

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2. Create an interface **Switchable** with a method `turnOn()`. Implement the interface in **Light** and **Fan** classes to display the device status.

```
1 interface Switchable {
2     void turnOn();
3 }
4 class Light implements Switchable {
5     public void turnOn() {
6         System.out.println(x: "Light is now ON");
7     }
8 }
9 class Fan implements Switchable {
10    public void turnOn() {
11        System.out.println(x: "Fan is now ON");
12    }
13 }
14 public class SwitchableDemo {
15     Run | Debug
16     public static void main(String[] args) {
17         Switchable s1 = new Light();
18         Switchable s2 = new Fan();
19         s1.turnOn();
20         s2.turnOn();
21     }
```

```
Light is now ON
Fan is now ON
```
